

No.ofFunctionalFeaturesIncludedinthe Solution

Date	3july2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5Marks

FunctionalFeaturesOverview

This document presents the functional features implemented in the EduGenie project. EduGenie is an AI-powered educational assistant that leverages Generative AI to provide intelligent learning support through question answering, concept explanation, quiz generation, educational text summarization, and personalized learning path recommendations. Each feature has been successfully implemented, tested, and integrated into the overall system, ensuring a smooth and interactive learning experience.

Functional Features Included

Name	Feature Description	Status	Module / Component	Marks	S.No	Feature Contribution
1	Question Answering responses.	Answers educational questions using Generative AI with contextual Gemini API	Intelligent	Done	5	
2		Simplifies complex topics into easy-Concept Explanation to-understand explanations.	LaMini-Flan-T5	Done	5	
3		Automatically generates topic-AI Quiz Generation based multiple-choice questions.	Gemini API	Done	5	
S.No	Feature Name	Description	Status	Module / Component	Marks	Feature Contribution
7	Interactive Web Interface	functionalities. Responsive HTML/CSS frontend for HTML, CSS, interaction.	Jinja2		Done	5 user
8	Error Handling & Validation	Handles invalid inputs and API gracefully.	Python	Utilities	Done	5 exceptions
9	Secure Configuration	Stores API keys and configuration environment variables.	.env using Configuration		Done	5
	AI Module Integration	Integrates multiple AI models into a AI Services unified educational platform.			Done	5

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	8
Additional Features	2
Features Tested & Verified	10

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Interactive Learning Dashboard
2	Backend / Logic	3	FastAPI APIs, Request Processing, Error Handling
3	AI & NLP Modules	4	Question Answering, Concept Explanation, Quiz Generation, Text Summarization
4	Recommendation System	1	Personalized Learning Path
5	Configuration & Security	1	Environment Variables, API Key Management